



CINEA/EMFAF/2021/3.1.2/03/SC04/SI2.881222

*Specific Contract 2021/3.1.2/03/SC04*

Hosting, maintenance and further development of the Regional Database for the  
Mediterranean and Black Seas (RDBFIS II)



MEDITS COORDINATION GROUP  
Kavala (Greece), 6<sup>th</sup> - 7<sup>th</sup> May 2025  
Fisheries Research Institute

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MEDITS database structure

[TA, TB, TC, TE, TL](#)

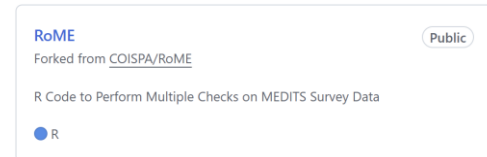
Integration within RDBFIS



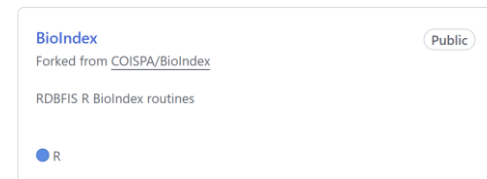
Validation scheme

[CSV Validator](#)

RoME (*R package*)



BioIndex (*R package*)



Status of MEDITS data upload

## Aggregated data

- Med&BS
- FDI (Tables A, B, G, H, I, J)
- AER
- GFCM/DCRF

**MEDITS** (TA, TB, TC)

MEDIAS (acoustics, pelagic trawl, CTDs)

## Detailed biological & landings data

### A straight-forward process to upload data



Step 1. Syntax check (validation scheme)

Step 2. Consistency check (quality control routines)

Step 3. Upload (update RDBFIS)

# RDBFIS: Status of MEDITS data upload



|           | BGR | CYP | ESP  | FRA | GRC  | HRV  | ITA   | MLT | ROU | SVN |
|-----------|-----|-----|------|-----|------|------|-------|-----|-----|-----|
| medits_ta | 621 | 440 | 3889 | 581 | 2294 | 1312 | 11752 | 809 | 322 | 48  |
| 2002      |     |     | 136  |     |      | 60   | 602   |     |     | 2   |
| 2003      |     |     | 140  |     | 173  | 59   | 603   |     |     | 2   |
| 2004      |     |     | 128  |     | 177  | 61   | 596   |     |     | 2   |
| 2005      |     | 25  | 123  |     | 178  | 59   | 644   | 45  |     | 2   |
| 2006      |     | 25  | 147  |     | 176  | 59   | 650   | 45  |     |     |
| 2007      |     | 25  | 164  |     |      | 61   | 630   | 46  |     | 4   |
| 2008      |     | 27  | 176  |     | 179  | 59   | 529   | 45  |     | 2   |
| 2009      |     | 27  | 169  |     |      | 60   | 536   | 45  |     | 2   |
| 2010      |     | 27  | 131  |     |      | 60   | 532   | 45  |     | 2   |
| 2011      |     | 26  | 170  |     |      | 61   | 534   | 44  |     | 2   |
| 2012      |     | 26  | 180  |     |      | 60   | 534   | 44  |     | 2   |
| 2013      |     | 25  | 199  |     | 66   | 59   | 593   | 45  |     | 2   |
| 2014      | 38  |     | 224  |     | 185  | 56   | 595   | 45  |     | 2   |
| 2015      | 38  | 26  | 233  |     |      | 65   | 592   | 45  |     | 2   |
| 2016      | 72  | 26  | 230  | 85  | 194  | 56   | 593   | 45  |     | 2   |
| 2017      | 80  | 26  | 228  | 90  |      | 61   | 537   | 45  |     | 2   |
| 2018      | 77  | 26  | 219  | 89  | 203  | 64   | 532   | 45  |     | 2   |
| 2019      | 80  | 25  | 226  | 94  | 195  | 61   | 603   | 45  | 82  | 3   |
| 2020      | 80  | 26  | 108  | 38  | 194  | 58   | 568   | 45  | 80  | 3   |
| 2021      | 80  | 26  | 272  | 89  | 186  | 89   | 712   | 45  | 80  | 4   |
| 2022      | 76  | 26  | 286  | 96  | 187  | 84   | 37    | 45  | 80  | 4   |

|           | BGR | CYP   | ESP    | FRA   | GRC   | HRV   | ITA    | MLT   | ROU | SVN  |
|-----------|-----|-------|--------|-------|-------|-------|--------|-------|-----|------|
| medits_tb | 509 | 13307 | 172178 | 29402 | 66909 | 36131 | 318740 | 19298 | 243 | 1251 |
| 2002      |     |       | 5127   |       |       | 1673  | 14425  |       |     | 47   |
| 2003      |     |       | 4838   |       | 4598  | 1648  | 15153  |       |     | 67   |
| 2004      |     |       | 4995   |       | 4631  | 1776  | 15347  |       |     | 58   |
| 2005      |     | 614   | 5066   |       | 4663  | 1676  | 16740  | 1042  |     | 44   |
| 2006      |     | 641   | 6005   |       | 4579  | 1615  | 16871  | 1172  |     |      |
| 2007      |     | 647   | 6689   |       |       | 1720  | 16198  | 1283  |     | 119  |
| 2008      |     | 574   | 5885   |       | 4477  | 1806  | 13707  | 1389  |     | 59   |
| 2009      |     | 738   | 6246   |       |       | 1647  | 14502  | 1392  |     | 58   |
| 2010      |     | 664   | 4713   |       |       | 1534  | 14262  | 1124  |     | 64   |
| 2011      |     | 721   | 5890   |       |       | 1418  | 15117  | 1225  |     | 57   |
| 2012      |     | 742   | 8259   |       |       | 1632  | 16049  | 1240  |     | 48   |
| 2013      |     | 710   | 9054   |       | 2134  | 1661  | 16841  | 1352  |     | 61   |
| 2014      | 28  |       | 11138  |       | 4674  | 1543  | 16315  | 1112  |     | 58   |
| 2015      | 31  | 843   | 9886   |       |       | 1833  | 16736  | 1165  |     | 66   |
| 2016      | 51  | 798   | 12191  | 4679  | 5436  | 1631  | 18187  | 1074  |     | 57   |
| 2017      | 59  | 833   | 10552  | 4629  |       | 1691  | 16781  |       |     | 43   |
| 2018      | 60  | 1010  | 11210  | 4959  | 5900  | 1734  | 16357  | 944   |     | 53   |
| 2019      | 67  | 1018  | 11796  | 5450  | 6623  | 1709  | 17078  | 1014  | 62  | 62   |
| 2020      | 71  | 1021  | 5367   | 1087  | 6758  | 1663  | 13505  | 833   | 55  | 59   |
| 2021      | 68  | 885   | 13601  | 3338  | 6493  | 2347  | 18354  | 910   | 63  | 67   |
| 2022      | 74  | 848   | 13670  | 5260  | 5943  | 2174  | 215    | 1027  | 63  | 104  |

|           | BGR  | CYP   | ESP    | FRA    | GRC    | HRV    | ITA     | ROU | SVN  |
|-----------|------|-------|--------|--------|--------|--------|---------|-----|------|
| medits_tc | 1902 | 59633 | 525709 | 129401 | 303914 | 185703 | 1188241 | 852 | 4504 |
| 2002      |      |       | 13310  |        |        | 9081   | 42666   |     | 265  |
| 2003      |      |       | 12495  |        | 22005  | 8863   | 46215   |     | 220  |
| 2004      |      |       | 11241  |        | 19208  | 9049   | 46439   |     | 120  |
| 2005      |      | 1345  | 11484  |        | 22564  | 7924   | 61679   |     | 401  |
| 2006      |      | 3256  | 15823  |        | 22159  | 7282   | 61123   |     |      |
| 2007      |      | 3509  | 36014  |        |        | 8463   | 62420   |     | 444  |
| 2008      |      | 2884  | 29994  |        | 22312  | 9081   | 49937   |     | 139  |
| 2009      |      | 3853  | 31448  |        |        | 7217   | 51476   |     | 150  |
| 2010      |      | 3098  | 17160  |        |        | 7082   | 54130   |     | 110  |
| 2011      |      | 3573  | 20775  |        |        | 6661   | 53127   |     | 188  |
| 2012      |      | 3623  | 19231  |        |        | 7680   | 57469   |     | 259  |
| 2013      |      | 3487  | 23058  |        | 8735   | 9592   | 61107   |     | 124  |
| 2014      | 88   |       | 27122  |        | 21876  | 9313   | 69450   |     | 259  |
| 2015      | 114  | 4612  | 26808  |        |        | 10297  | 60798   |     | 279  |
| 2016      | 179  | 3921  | 30646  | 21510  | 25395  | 8990   | 71773   |     | 173  |
| 2017      | 141  | 4506  | 27726  | 18856  |        | 9907   | 68587   |     | 103  |
| 2018      | 132  | 4324  | 27101  | 19920  | 31710  | 8500   | 63635   |     | 199  |
| 2019      | 209  | 3895  | 33644  | 20390  | 28570  | 9263   | 69827   | 192 | 286  |
| 2020      | 351  | 4154  | 22638  | 8236   | 29396  | 8993   | 57727   | 215 | 228  |
| 2021      | 360  | 3018  | 43408  | 17346  | 24525  | 11029  | 75945   | 239 | 241  |
| 2022      | 328  | 2575  | 44583  | 23143  | 25459  | 11436  | 2711    | 206 | 316  |



### MEDITS integration within RDBFIS: RoME

Under the scope of the RDBFIS and QualiTrain projects, the RoME and BioIndex packages were reviewed and modified to address issues identified during bilateral meetings and trainings with Member States and to optimize software performance.

The COISPA, in collaboration with UOA and HCMR worked together to address the generating RoME outputs as a consolidated ZIP file, simplifying the download process for RDBFIS users.

A key issue raised during the bilateral meetings was the inconsistency in MEDITS data, where species records were present in the TC table but missing from the TB. Although the RoME checks accurately flagged this error in both log files and error reports, the RDBFIS team further refined the handling of this issue to ensure that such data discrepancies would interrupt the import process into the RDBFIS database, enhancing data validation and integrity.



## MEDITS integration within RDBFIS: BioIndex

A significant advancement within the RDBFIS project is the integration of the BioIndex tool for the analysis of MEDITS survey data. BioIndex is a dedicated R package designed to streamline the processing and analysis of fisheries survey datasets, particularly those adhering to the MEDITS data exchange format. The tool offers over 30 specialized functions supporting time series analyses, spatial evaluations, and demographic assessments across multiple geographical scales. Analyses can be conducted at both GSA level and at the country level within a GSA.

*A user-friendly graphical interface has been developed to streamline the configuration of BioIndex analyses. The interface allows users to set key parameters such as spatial filters (country, GSA, depth strata), reference species, time series ranges, and species-specific size thresholds for life stage classification. The analysis outputs are stored in ZIP archives. The package produces both CSV tables for further statistical use and a range of graphical outputs—including boxplots, bubble plots, heatmaps, and spatial grids—to enhance data interpretation.*



## MEDITS integration within RDBFIS: BioIndex

The statistical and biological indicators computed by BioIndex include, among the other:

- Abundance indices (number/km<sup>2</sup>)
- Biomass indices (kg/km<sup>2</sup>)
- Mean Individual Weight (MIW).
- Length-based indicators (L50-L95)
- Life stage-specific indices, differentiating between recruits and spawners to provide insights into stock renewal potential and recruitment success.
- Sex ratio assessments
- Inverse of the coefficient of variation, which describes the stability of abundance indices over time.

BioIndex also supports spatial analyses through the GFCM 0.5° x 0.5° statistical grid, allowing users to map species distributions and detect spatial patterns in abundance, biomass, sex ratio and other indicators



## MEDITS integration within RDBFIS: supporting materials

[https://github.com/medbsrdb/RDBFIS-materials/blob/main/RDBFIS-II/Presentations/Training/2nd%20Training/MEDITS\\_RoME.pptx](https://github.com/medbsrdb/RDBFIS-materials/blob/main/RDBFIS-II/Presentations/Training/2nd%20Training/MEDITS_RoME.pptx)

<https://github.com/medbsrdb/RDBFIS-materials/blob/main/RDBFIS-II/Presentations/Training/2nd%20Training/BioIndex.pptx>

[https://github.com/medbsrdb/RDBFIS-materials/blob/main/RDBFIS-II/Presentations/Training/2nd%20Training/BioIndex\\_Technicaldocument.pdf](https://github.com/medbsrdb/RDBFIS-materials/blob/main/RDBFIS-II/Presentations/Training/2nd%20Training/BioIndex_Technicaldocument.pdf)



## MEDITS integration within RDBFIS: Suggestions

Several comments/suggestions have been raised that could positively contribute to the harmonization of the Med&BS, FDI, GFCM/DCRF and AER datacall databases used by Member States for data analysis and reporting obligations. These comments primarily concern:

- Standardized codification systems, ensuring consistency across datasets (JRC, GFCM, RDBFIS)
- Ensure strict compliance with the definitions specified in the data call annexes, including precise requirements for decimal places, maximum haul depth, and other key parameters (separate meetings with JRC and GFCM are required to address these cases)
- Provide the possibility to delete existing records
- Summary statistics will serve as a valuable and informative addition to the application dashboard
- Enable support for importing .xlsx files in the data import process
- Optimize and streamline consistency checks to reduce processing time
- Several issues were identified concerning discrepancies in species scientific names across different species lists (e.g. MEDITS, FAO, WoRMS).

# Establishment of the Regional Database for the Mediterranean and Black Seas

Acronym: Med&BS RDBFIS-III

Under FRAMEWORK CONTRACT – EASME/EMFF/2020/OP/0021

Framework Contract for the provision of scientific advice  
for the Mediterranean and the Black Seas

Specific Contract CINEA/EMFAF/2024/3.1/01/SC08

Contractor: Hellenic Centre for Marine Research (HCMR)

Specific Contract Project Coordinator: **Dr. Irida Maina (HCMR)**

Scientific Information Technology Coordinator: Stefanos Kavadas

Duration: 13 months and 21 days

Project Start: April 1<sup>st</sup> 2025

## Project structure

### Aim

- Continue the hosting, maintenance, fine-tuning and further development of RDBFIS;
- Finalize the population of RDBFIS with data (from existing data calls) and agree with the relevant actors on an all-encompassing data call for the future;
- Work towards using RDBFIS for submission to the main data calls and reporting obligations;
- Provide support to the users of RDBFIS and to ensure the migration to the permanent host once a decision is taken.

**Sub-Task 0.1** – Project management

**Sub-Task 0.2** – Meetings with CINEA/DG MARE

**Sub-Task 0.3** – Reporting activities

### Task 0 – Project Management and Coordination

**Task 1** Provision of a temporary solution for hosting and maintenance of RDBFIS

**Task 2** Population of RDBFIS with MS data and future data call

**Task 3** Submission of data to end users through RDBFIS

**Task 4** Fine-tuning and further development

**Task 5** Provision of support to the users

**Task 6** Advise on way forward, in terms of improving the RDB for the region

**Task 7** Participation in meetings, where the work of this contract needs to be disseminated



**Task 8:** Prepare a hand over file to facilitate the transition of RDBFIS to DG MARE, RCG Med&BS and the final host

## **Task 2: Population of RDBFIS with MS data and future data call**

**Coordinators:** Stefanos Kavadas, Irida Maina-HCMR

**Duration:** 11 months (from month 1 to 11, April 2025-February 2026)

Subtasks:

i) Finalize the population of the system with data and to update with the latest data sets. Data calls already being launched or ongoing:

a) Complete the upload process for detailed biological data

b) MEDITS data from 1994 – 2001

c) Complete the upload process for MEDIAS acoustics, pelagic trawl and CTD data

if the need arises for a new data call, the contractor will draft the technical elements, to be addressed to the relevant MS and actors of the region, such as RFMOs

ii) Propose a data call for the future years. A recurrent (annual) data call should be agreed with the relevant actors, ensuring that all the data needs of the main end users are fulfilled. The contractor will draft a data call and consult with interested parties and stakeholders (relevant MS, RCG Med&BS, DG MARE, JRC, STECF, GFCM etc.), before finalizing it.

iii) Monthly fishing effort data by GSA, gear and length category will be requested from the MS to support the advanced statistic component (will be developed in Task 4, Activity 10 - stat4RDBFIS software Package development).

## Med&BS RDBFIS-III: Fine-tuning and further development of RDBFIS (TASK 4)

**Coordinators:** Stefanos Kavadas, Irida Maina (HCMR), Ioannis Chamodrakas (UA)

**Partners involved:** All

**Duration:** 13 months (from month 1 to 13, April 2025- April 2026)

**Objective:** Under this task, an attempt will be made to complete ongoing areas of work from the previous study in need of further improvement, based on the experience gained, exchanges with users, namely the Med&BS MS and main end users of the region and change requests from DG MARE.

### **12 activities will be carried out to support the following objectives:**

- (1) database maintenance and restructuring,
- (2) enhancement and optimization of the existing programming code,
- (3) common raising procedures for the Med&BS biological data,
- (4) development of new software packages,
- (5) integration within RDBFIS of the existing sampling optimization package,
- (6) fleet analysis updates and improvements on FDI spatial checks,
- (7) integration within RDBFIS of the outcomes of MAPAFISH project,
- (8) Fine-tuning.

### Activity 2: Update/revision of existing data quality tools (objective 2)

Coordinators: Isabella Bitetto (COISPA), Walter Zupa (COISPA)

Duration: 4 months (from month 2 to 5, May 2025- August 2025)

- The present activity aims at further improving the functionalities of the **RoME** and **RDBqc** quality check tools, as well as the **BioIndex** data analysis tool, by improving the already existing functions, where needed, and implementing new ad hoc functions, where missing.
- A key objective of this activity is to address issues identified during earlier projects and training sessions by leveraging user feedback and insights gained from practical applications of these tools.
- The focus will be on enhancing existing features and introducing new functionalities to meet evolving requirements.



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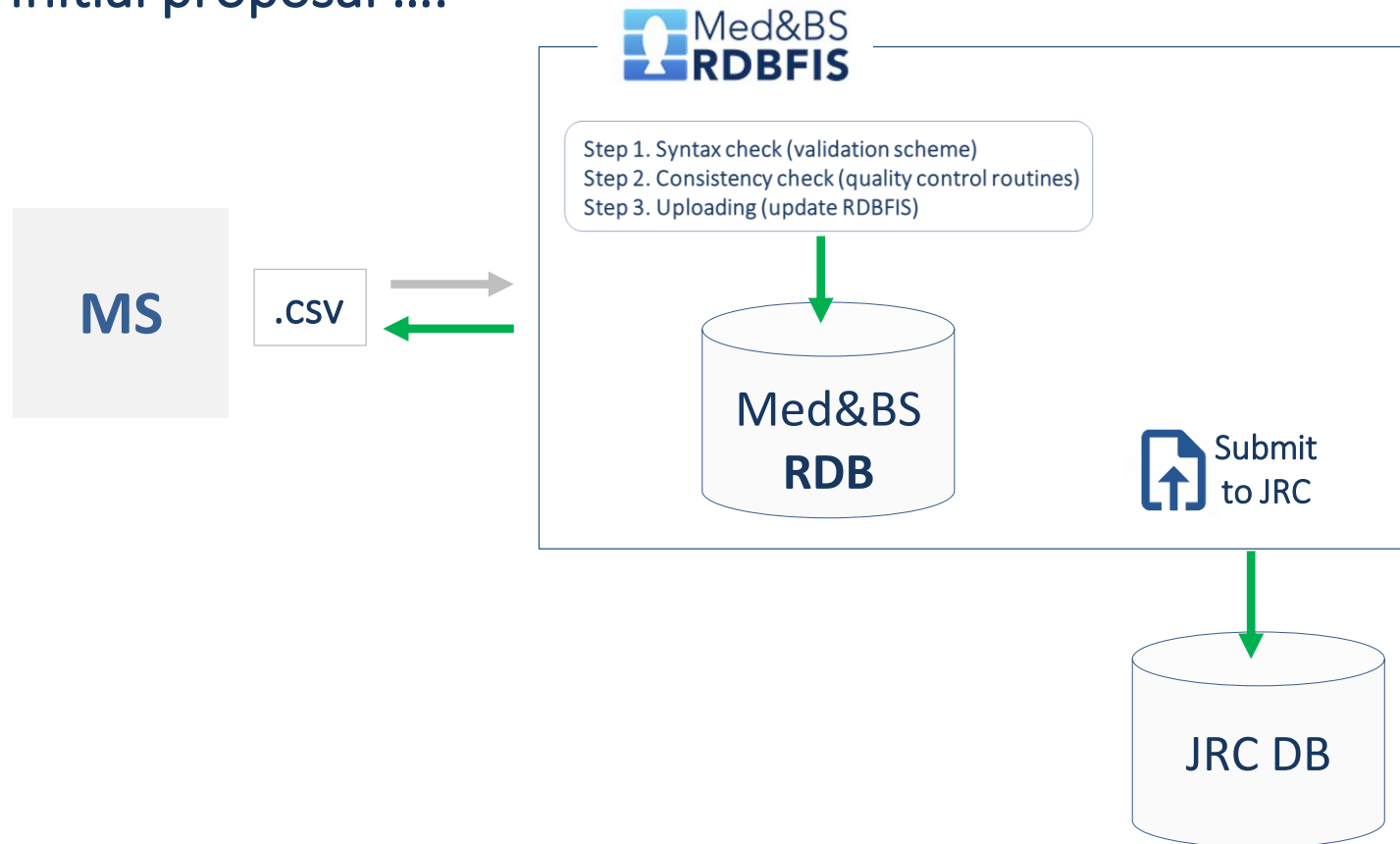
EASME/EMFF/2020/OP/0021

Specific Contract CINEA/EMFAF/2024/3.1/01/SC08

Establishment of the Regional Database for the Mediterranean and  
Black Seas (RDBFIS-III)



Initial proposal ....



... **but:** a) there are commission policy issues regarding data,  
b) is a much bigger task to alter the stringent institutional security bureaucracy

During the project implementation, twelve activities will be carried out to support the following eight objectives:

O1. database maintenance and restructuring

O2. enhancement and optimization of the existing programming code (Update/revision of existing data quality tools, tools4MCDA, spatial visualization, Improvement and further development of the existing validation scheme and consistency checks)

O3. common raising procedures for the Med&BS biological data

O4. development of new software packages (Development of data quality tools: MEDIAS, stat4RDBFIS software Package development)

O5. integration within RDBFIS of the existing sampling optimization package

O6. fleet analysis updates and improvements on FDI spatial checks

O7. integration within RDBFIS of the outcomes of MAPAFISH project

O8. Fine-tuning